

Day 4

Task 1:

Day 3 workflow was converted into a complete scikit-learn Pipeline containing the six engineered features, preprocessing and the model. The same stratified 70% train / 10% development / 20% untouched hold-out split from Days 1–3 was retained. Randomness was controlled with `random_state=42`, and the notebook records the Python, pandas, NumPy and scikit-learn versions.

Task 2:

Performed controlled `RandomizedSearchCV` tuning for Logistic Regression, Random Forest and `HistGradientBoosting`. Each search used 3-fold `StratifiedKFold` cross-validation, precision as the primary optimization metric, `n_jobs=-1` and a fixed random state. The search spaces covered regularization strength for Logistic Regression, tree size and sampling parameters for Random Forest, and learning rate, iterations, tree structure and L2 regularization for `HistGradientBoosting`. The best parameters and corresponding cross-validated precision are recorded directly from the search results.

Task 3:

Examined regularization and learning behaviour in two ways. First, a Logistic Regression C sweep compared training and validation precision across different regularization strengths. Second, a learning curve was generated for the tuned `HistGradientBoosting` model, comparing training and validation precision at increasing training-set sizes. The train-validation gap was then used to determine whether the model showed meaningful variance or bias and what direction of improvement would be appropriate, such as stronger regularization, simpler tree structure or additional data.

Task 4:

Evaluated the probability outputs of the best tuned candidate before and after sigmoid calibration using calibration curves and Brier score. The calibrated probabilities were then evaluated on the development set to select the operating classification threshold. Because precision is the business priority, the threshold selection maximizes F1 while maintaining the Day 3 `HistGradientBoosting` precision reference of 0.7764 when a feasible threshold exists. Also compared several thresholds and showed their effects on precision, recall, F1 and the confusion-matrix counts.

Task 5:

After model selection, calibration and threshold selection were complete, the final pipeline was refitted using the training and development data together. The hold-out test set remained untouched until this final evaluation. The final notebook reports accuracy, precision, recall, F1, ROC AUC, PR AUC, Brier score and TN/FP/FN/TP, together with ROC, precision-recall, calibration and confusion-matrix plots. Finally, the complete calibrated pipeline and its frozen decision threshold are saved together as a `.joblib` artifact.

Final Outcome:

Day 4 takes the strongest candidates from Day 3 and turns them into a controlled model-selection workflow. The important distinction is that tuning and threshold selection are performed without using the final hold-out set. This keeps the final test evaluation as an unbiased estimate of how the finished pipeline performs on unseen data. The saved artifact contains the preprocessing, feature engineering, calibrated model and operating threshold so the same decision process can be reproduced during inference.